Project 3: Vannilas Supermarket Sales and Inventory Analysis

Project Title

Vannilas Supermarket Sales and Inventory Analysis

Background

Vannilas Supermarket is one of the busiest retail businesses serving customers in Kakuma Town. The supermarket sells groceries, beverages, household items, personal care products, electronics, and other consumer goods.

Over the last three years, the supermarket has recorded thousands of transactions. Management wants to better understand sales performance, customer purchasing patterns, and inventory movement to improve profitability and reduce stock shortages.

As a Data Analyst, you have been hired to analyze supermarket sales data and provide recommendations to management.

Project Objective

Analyze sales and inventory data to identify trends, top-performing products, customer purchasing behavior, and inventory challenges.

Business Questions

1. What was the total revenue generated?
2. Which products generated the most sales?
3. Which product categories are most profitable?
4. Which months recorded the highest sales?
5. Which branches or departments perform best?
6. Which products sell slowly?
7. Which products experience stock shortages?
8. Which payment methods are most commonly used?
9. What are the top-selling products?
10. What recommendations can increase sales and improve inventory management?

Dashboard Requirements

KPI Cards

- Total Revenue
- Total Transactions
- Total Products Sold
- Average Transaction Value
- Gross Profit
- Inventory Turnover Rate

Visualizations

- Revenue by Product Category
- Monthly Sales Trend
- Revenue by Payment Method
- Top 10 Best-Selling Products
- Lowest-Selling Products
- Revenue by Department
- Inventory Levels by Category
- Stock-Out Analysis
- Profit by Category
- Customer Purchase Trend

Filters / Slicers

- Year
- Month
- Product Category
- Product Name
- Payment Method
- Department

Dataset Information

Dataset Name: Vannilas_Supermarket_Sales_Inventory.xlsx

Records: 15,000+

Product Categories

- Groceries
- Beverages

- Household Items
- Personal Care
- Electronics
- Stationery
- Baby Products
- Clothing